Australian Journal of Earth Sciences (2009) **56**, (201–212)

Paleomagnetic evidence for periods of intense oxidative weathering, McKinnons mine, Cobar, New South Wales

M. L. SMITH[1]*, B. J. PILLANS[1] AND K. G. McQUEEN[2]

[1]*Cooperative Research Centre for Landscape Evolution and Mineral Exploration (CRC LEME), Research School of Earth Sciences, Australian National University, Canberra, ACT 0200, Australia.*
[2]*Cooperative Research Centre for Landscape Evolution and Mineral Exploration (CRC LEME), Department of Earth and Marine Sciences, Australian National University, Canberra, ACT 0200, Australia.*

The McKinnons gold mine, ~35 km southwest of Cobar in New South Wales is a non-operational opencut mine with extensive regolith profile development down to a depth of ~80 m. Twenty-three oriented paleomagnetic samples, from saprolite at different levels through the profile, yielded two well-defined paleomagnetic poles, one of Paleocene age (60 ± 10 Ma), and the other of Miocene age (15 ± 4 Ma). These reflect two periods of remagnetisation associated with deep, intense, oxidative weathering under humid conditions. They provide important time constraints for the history of climate change, weathering and landscape evolution of the Cobar region.

KEY WORDS: Cobar, McKinnons mine, New South Wales, paleomagnetic poles, regolith, weathering.

Downloaded By: [University Of Western Australia] At: 12:50 25 March 2009

INTRODUCTION

Time constraints are critical for a proper understanding of landscape evolution. Specific weathering and landscape evolution events are generally difficult to constrain, and an age range provided by stratigraphy may be the only temporal constraint that can be obtained (Pillans 2005). Generally speaking, regolith-dating methods, such as radiocarbon, luminescence, U-series and cosmogenic nuclides, are restricted to dating events less than *ca* 5 Ma (Pillans 1998). More ancient weathering events in regolith materials can be dated by methods such as argon dating (K–Ar and $^{40}Ar/^{39}Ar$) (Vasconcelos *et al.* 1992; Dammer *et al.* 1996), paleomagnetism (Idnurm & Senior 1978; O'Sullivan *et al.* 2000), $\delta^{18}O$ of clays (Bird & Chivas 1989) and (U–Th)/He (Pidgeon *et al.* 2004; Heim *et al.* 2006).

Different geochronological techniques date different weathering and landscape-forming events. For example, paleomagnetism dates the terminal, dehydration stages of intense, chemical weathering or increased seasonality (Idnurm & Schmidt 1986), while $\delta^{18}O$ clay dating records periods of mineral formation under more generally humid conditions (Bird 1988). Thus, different methods may be combined to give a range of different ages for different materials and processes from the same profile.

Thick regolith profiles are developed in the Cobar region and have altered the surface expressions of ore bodies at the Elura, CSA, Peak, New Cobar and McKinnons mines. However, apart from the recent study by McQueen *et al.* (2007), little is known about the age of the regolith in the Cobar region. Here we report the results of a paleomagnetic dating study of the weathering profile at the McKinnons gold mine, 35 km southwest of Cobar.

PALEOMAGNETIC DATING OF REGOLITH IN AUSTRALIA

Principles

The most important mineral for paleomagnetic dating of ferruginous weathered profiles in Australia is hematite, which is chemically stable within the regolith (Idnurm & Schmidt 1986). The characteristic remanence carried by hematite is a chemical remanent magnetisation (CRM), acquired during weathering of regolith materials under oxidising conditions (Butler 1992). An important characteristic of such CRMs is that they are acquired over sufficiently long periods of time (>10 ka) to average out secular variation of the Earth's magnetic field. They can thus be used to determine paleomagnetic pole positions. Furthermore, extended or intense weathering generally results in the destruction of any prior remanent magnetisation. Thus, paleomagnetic ages

*Corresponding author and present address: Newmont Asia Pacific, Tanami Operations, PO Box 8020, Alice Springs, NT 0871, Australia (martin.smith@newmont.com).

ISSN 0812-0099 print/ISSN 1440-0952 online © 2009 Geological Society of Australia
DOI: 10.1080/08120090802547033

from regolith materials are related to weathering events, rather than the time of rock formation.

In our experience, the preservation of a stable magnetic remanence in regolith is favoured when hematite is formed in intensely weathered saprolite, below the zone of modern soil formation, and below any transported cover. In their review of paleomagnetic studies of regolith profiles, Idnurm & Schmidt (1986) postulated that two key environmental conditions for the formation of deep, chemically weathered profiles are a humid climate, favourable to chemical weathering, and a low-relief topography. Idnurm & Schmidt (1986) also outlined the necessary factors for the development of stable remanence, and concluded that these factors preclude continuous magnetisation. Rather, they appear to be best met in dehydration reactions during the terminal stages of intense weathering, or shortly after the intense weathering has ceased. Thus, while magnetic minerals may be precipitated, altered, dissolved and re-precipitated during the formation of a weathered profile, the age of acquisition is likely to correspond to the latter stages of active weathering.

Paleomagnetic weathering ages are calculated by comparing weathering poles with the Australian Apparent Polar Wander Path (AAPWP). In this study we use the AAPWP of Schmidt & Clark (2000) as our reference path, which includes the revised Late Mesozoic to Cenozoic poles of Idnurm (1985, 1994).

Previous work

One of the first studies to use the AAPWP for dating weathering events in Australia was by Schmidt & Embleton (1976), who reported paleomagnetic results from Late Paleozoic and Mesozoic rocks in the Perth Basin and adjacent areas of Western Australia. They identified a stable high-temperature magnetisation, which was similar in direction for all rocks, regardless of age, and which they interpreted as a blanket remagnetisation resulting from a period of regional lateritisation. The age of the weathering-induced remagnetisation was estimated to be Late Oligocene to Early Miocene by comparison with the AAPWP of McElhinny & Embleton (1974). More recent revisions of the AAPWP (Idnurm 1985, 1994) indicate a Late Miocene or Pliocene age (6 ± 4 Ma). Schmidt *et al.* (1976) studied two sites in South Australia—a subbasaltic weathering profile on Kangaroo Island and red clays in the Springfield Basin, near the Flinders Ranges—and found similar paleomagnetic poles to those from the Perth Basin, with similarity to a secondary (overprint) pole from the Montejinni Limestone in the Northern Territory (Luck 1970). They concluded that the poles may reflect a synchronous weathering event that affected a large area of Australia.

Several studies have yielded ages ranging from latest Cretaceous through to Neogene at various sites across the continent (Pillans 1977; Idnurm & Senior 1978; Bishop *et al.* 1982; Schmidt *et al.* 1982; Schmidt & Ollier 1988; Nott *et al.* 1991; Acton & Kettles 1996). Idnurm & Senior (1978) determined a mid-Paleocene paleomagnetic age for the widespread, ferruginous Morney weathering profile in the Eromanga and Surat Basins in Queensland. Subsequent work by Pillans (2002, 2005) at a number of sites has yielded similar mid-Paleocene ages suggesting that these represent another continent-wide episode of intense, deep oxidative weathering in Australia.

Other studies have reported pre-Late Cretaceous paleomagnetic weathering ages (summarised in Pillans 2005). In the Cobar region, at the New Cobar mine, 3 km southeast of Cobar, a Jurassic age has been obtained by Pillans (*in* McQueen *et al.* 2007). At the Northparkes copper–gold mine, near Parkes in central New South Wales Pillans *et al.* (1999) reported mid-Carboniferous weathering ages. When combined with apatite fission-track data (O'Sullivan *et al.* 2000), the paleomagnetic data showed that the landscape at Northparkes had undergone multiple phases of stability and weathering as well as kilometre-scale burial and erosion.

REGIONAL GEOLOGY

The Early Devonian Cobar Basin (Figure 1) located in central western New South Wales, forms part of the Central Subprovince of the Lachlan Orogen. The basin formed in a backarc, intracratonic basin setting, with coeval deep-water troughs and flanking shallow-water shelves (Glen 1987; Glen *et al.* 1996). It was inverted and structurally deformed in the Middle Devonian to Carboniferous. The basin fill (Nurri Group and Amphitheatre Group) consists of sedimentary rocks, mainly of siliciclastic facies. Trough sediments deposited during initial basin rifting included turbidites, very minor volcanic rocks and, nearby, minor limestones (Nurri Group, lower Amphitheatre Group and Biddabirra Formation). Later infilling deposits were mainly deep-water turbidites of the Biddabirra Formation and upper Amphitheatre Group (Glen *et al.* 1985; Byrnes 1993). Older, Paleozoic rocks are dominantly metasedimentary sequences, including the Girilambone and Wagga Groups, and intruding Silurian granites. The Lower Devonian Cobar Supergroup is partly overlain by Lower to Upper Devonian fluvial sediments of the Mulga Downs Group (Byrnes 1993; Glen *et al.* 1996).

Sediments of the Mesozoic Eromanga and Surat Basins overlie the Paleozoic rocks north of the Cobar region, and the Cenozoic Murray Basin onlaps to the southwest (Figure 1). The Eromanga Basin contains major transgressive–regressive sequences spanning the Jurassic to mid-Cretaceous. The last marine incursion into the Eromanga–Surat Basin ended in the Albian (*ca* 110 Ma), and since that time the region is inferred to have been subaerially exposed (Frakes *et al.* 1987). The Murray Basin consists of mostly non-marine sands, silts and clays with some poorly lithified marine sedimentary rocks in the south. These are overlain by Quaternary dunefields and fluvio-lacustrine sediments (Brown & Stephenson 1991), the character of which generally reflects the change to a more arid climate and related landscape processes.

Rocks in the Cobar Basin are deeply and intensely weathered with leached outcrops, residual soils, and alluvial and colluvial cover of varying thickness (McQueen *et al.* 2007). The region has generally low relief, with more resistant, locally silicified lithologies

Downloaded By: [University Of Western Australia] At: 12:50 25 March 2009

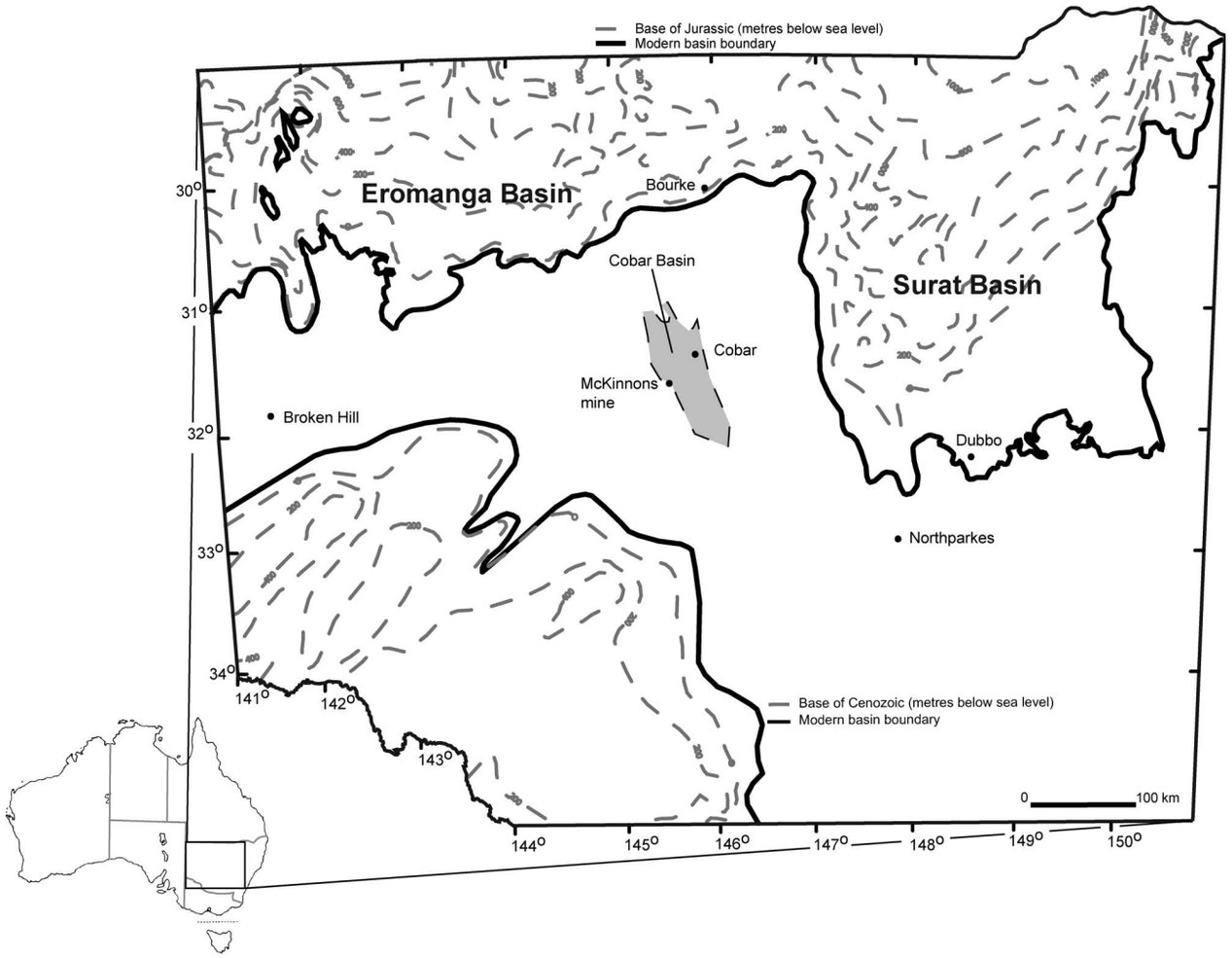


Figure 1 Mapped extent of the Eromanga–Surat Basin system and Murray Basin in New South Wales, with depth to the base of the Jurassic (Eromanga–Surat) and Cenozoic (Murray) sequences (compiled from Bembrick 1975; Hawke *et al.* 1975). Approximate position of Cobar Basin shown.

forming low ridges. A veneer of lag gravel is widespread. Thick alluvial and colluvial material rests on truncated weathering profiles along more densely vegetated drainage lines and associated flood plains. Soils in the region are dominantly silty-loam red earths, showing little profile development and containing a significant eolian component (Cohen *et al.* 1996; Tate *et al.* 2007).

McKINNONS MINE

The McKinnons mine is located near the southwestern margin of the Cobar Basin (Figure 1) near the axial trace of the Nullawarra Anticline, a major northwest (F_1)-trending fold hinge (Glen 1987). It is hosted by low-grade metasedimentary rocks of the lower Amphitheatre Group (Early Devonian), which includes turbiditic siltstones and sandstones and limestone lenses. The McKinnons deposit is in an erosional setting of subdued relief with undulating low rises (<20 m). Progressive erosional stripping and downward development of the weathering profile resulted from gentle northward tilting of this part of the Cobar Basin during the Early Cenozoic (Duk-Rodkin *et al.* 2004). This is also evidenced by apparently decreasing depths of erosion to the north.

Weathering at the McKinnons mine extends to more than 80 m depth (Figure 2), with the economic mineralisation only found in the saprolite (Marshall & Scott 1999; McQueen *et al.* 2005). A detailed study of the geology and structure of the McKinnons deposit can be found in Elliott *et al.* (1998). Representative paleomagnetic samples were taken from several areas of the McKinnons openpit (Figure 3), with most from highly ferruginous saprolite which is known to yield more reliable paleomagnetic results.

Samples

At the base of the pit, the oxidation (weathering) front is irregular and largely controlled by joints, shear zones and other fractures. Samples taken from this area of the pit are characterised by grey to greenish and dark-red to orange mottles, a subhorizontal bedding fabric with iron staining and vertical quartz-veins-filling cleavage, up to 1 m from the faults (samples MCK21–MCK23: Figure 2).

Downloaded By: [University Of Western Australia] At: 12:50 25 March 2009

Downloaded By: [University Of Western Australia] At: 12:50 25 March 2009



Figure 2 McKinnons mine regolith (looking south). The depth of the pit (ground surface to water level) is ~180 m.

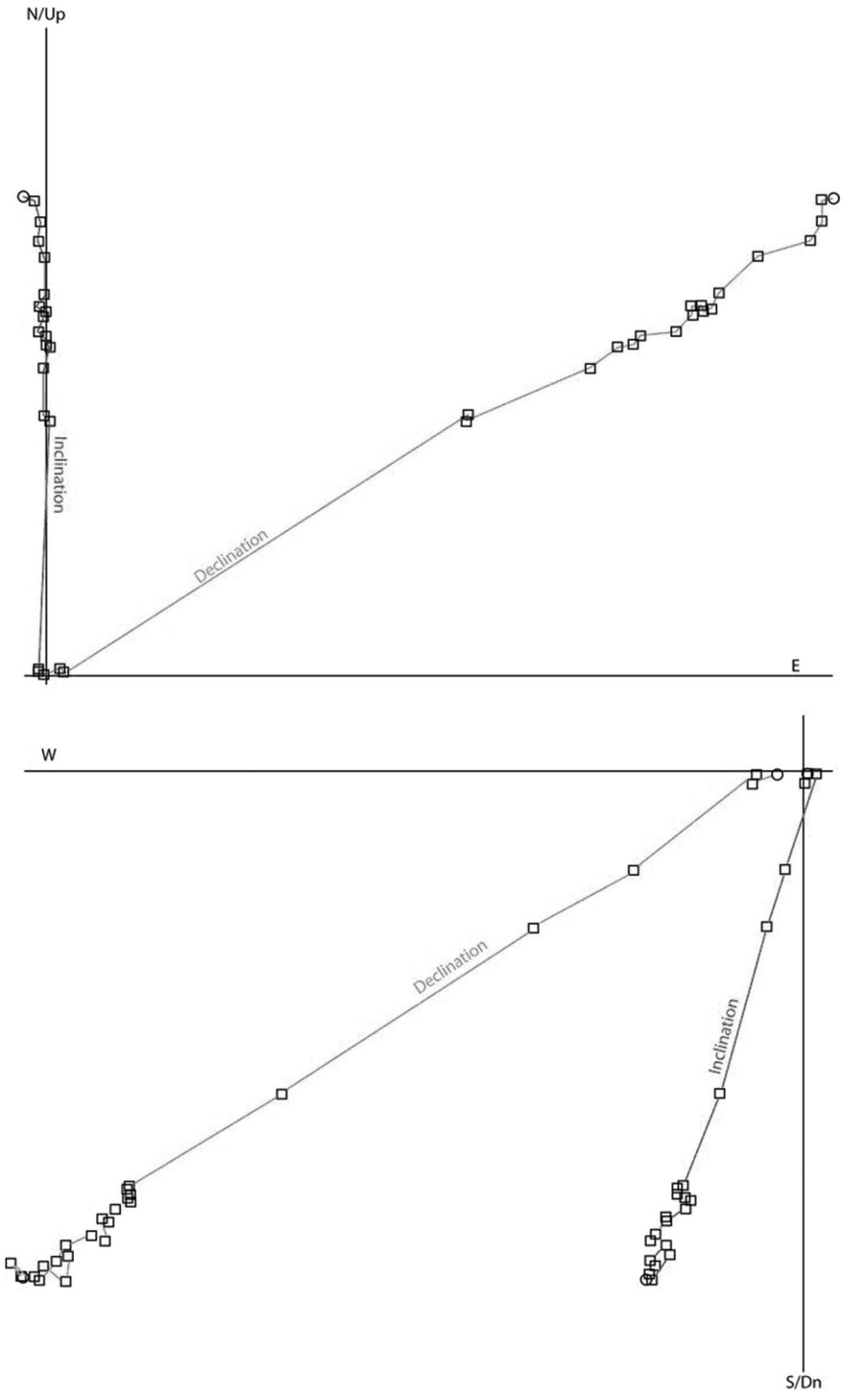


Figure 3 Representative Zjiderveld plots of thermal demagnetisation spectra for McKinnons paleomagnetic samples. Top, normal ChRM; bottom, reversed.

The main (unmined) sulfide lens is also partly weathered, with a green to yellow clay-rich coating, up to 1 mm thick on the exposed surfaces. The smaller, subsidiary sulfide lenses are dark-grey to black, in fine-grained, massive sandstone with a silicified, blocky outcrop pattern. This material is orthogonally jointed with infilling iron oxides. The rims of the lenses are characterised by more intensively oxidised material than the immediately adjacent parts of the profile. Around the sulfide lenses, iron oxidation and other weathering is still strongly controlled by shear zones, and has resulted in pinkish coloration of the surrounding saprock.

The bleached saprolite zone is the most prominent part of the regolith profile at the McKinnons mine (Figure 2). As with all other parts of the profile it is heavily silicified, with a major clay component. Macroscopic features such as jointing have been preserved, but bedding and other primary sedimentary structures have been largely destroyed where mineralisation is

Downloaded By: [University Of Western Australia] At: 12:50 25 March 2009

present. The bleached saprolite is generally friable or clay-rich, and unsuitable for paleomagnetic work. However, there are oxidised patches with concentric bands of pink, purple, yellow and red material, similar in appearance to liesegang banding, from which blocks were taken. Samples were also taken from the top of the bleached saprolite zone (samples MCK01–MCK04, MCK19–MCK20: Figure 2). These came from parts of the profile with concentric iron banding, which appears more intensely ferruginised than oxidised patches deeper in the bleached saprolite.

An intensely weathered and oxidised zone lies between the bleached zone and overlying mottled zone, from about 15 to 30 m depth (Figure 2). There is a major north–south shear associated with this heavily oxidised, bright red, yellow and orange material. It is generally blocky, and heavily jointed. In the southwestern corner of the McKinnons mine, there is a small pit, which exposes the transition from the bleached saprolite zone to the more ferruginous, upper zones. Material here consists of red, silicified and strongly fractured rock associated with a shear zone. There is iron oxide zoning, generally with a red core, orange transition zone and yellow-brown margin concentrated along and parallel to fractures. Eight samples were collected from this area (samples MCK05–MCK07, MCK13–MCK18: Figure 2).

The mottled zone at the McKinnons pit (Figure 2) contains two types of mottling. The upper part has mottles (10–100 cm in width) with both nodular and apparently bedding-parallel iron oxide concentrations. The latter include thin, resistant iron oxide material alternating with slightly thicker, less resistant iron oxides, which appear to pick out compositional banding in the parent rock. An area of mega-mottling immediately underlies this mottling. The mega-mottles consist of large (1–3 m wide), irregular patches of iron oxide-rich material separated by varying thicknesses of bleached white or orange, clay-rich material. Samples were collected from both the mottles and mega-mottles (samples MCK08–MCK11: Figure 2).

A veneer of red soil, up to 70 cm thick, overlies the mottled zone (Figure 2). In some places a thin stone-line is developed at the interface of the soil and saprolite. In the northeast corner and east wall of the pit there is a ~65 m-wide oblique section through a channel-fill deposit, which contains rounded and poorly sorted debris of generally coarse, mixed lithologies (quartz, siltstones and sandstones). The channel boundary is characterised a thick, calcareous layer. Mottling is strongly developed in the channel sediments.

Methods

Sampled blocks were oriented *in situ*, using either a sun compass or a Brunton compass, as described in Butler (1992). Several standard paleomagnetic cubes (~8 cm^3)

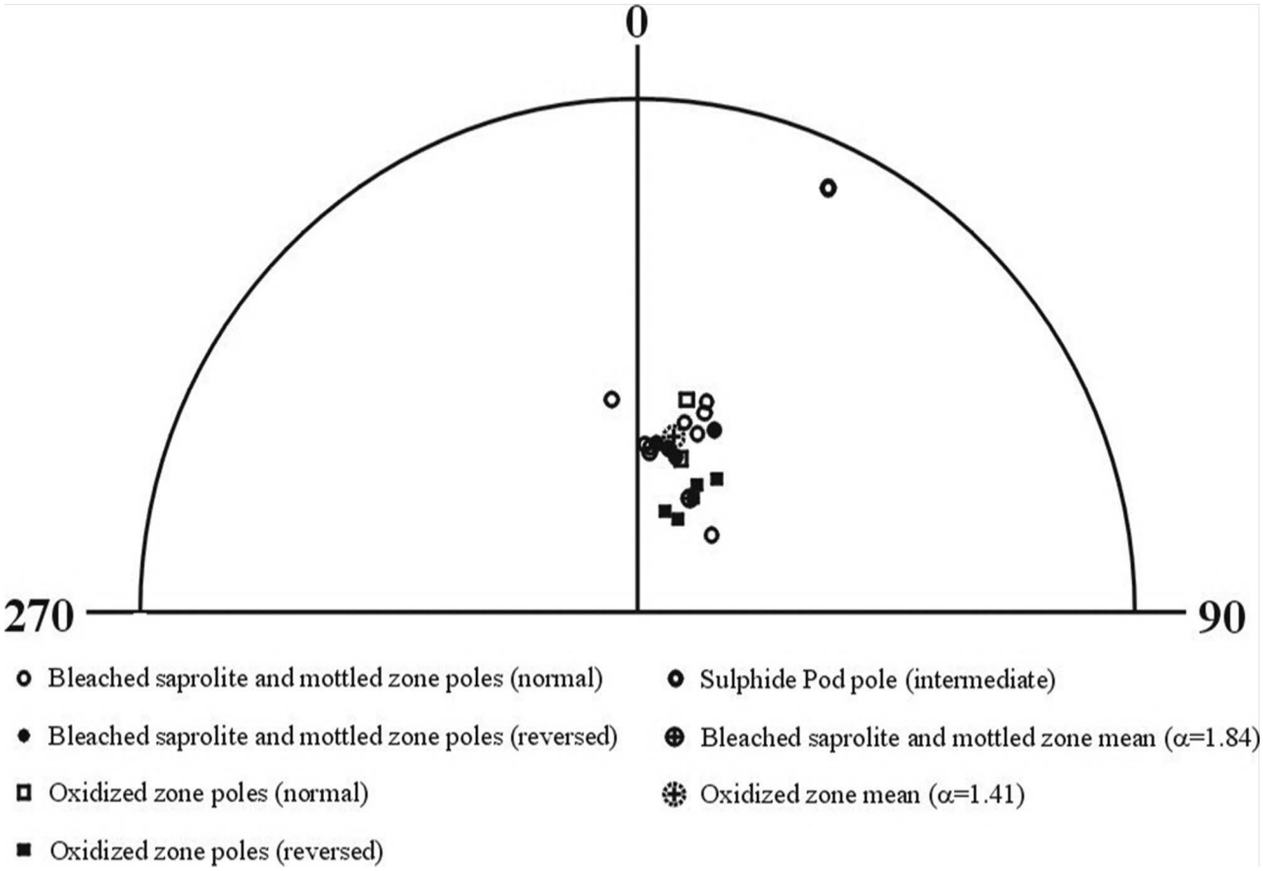


Figure 4 Stereoplot of ChRM for McKinnons samples. Two populations are evident in this plot, as represented by squares and circles.

Downloaded By: [University Of Western Australia] At: 12:50 25 March 2009

Downloaded By: [University Of Western Australia] At: 12:50 25 March 2009

were cut from each sample, and the natural remanent magnetism (NRM) of each specimen was measured on an ScT cryogenic magnetometer at the Australian National University Black Mountain Paleomagnetic Laboratory. Bulk susceptibility (χ) of each specimen was measured on a Digico susceptibility bridge. Following NRM measurements, specimens were subjected to stepwise thermal demagnetisation as outlined in Idnurm & Senior (1978). Declination, inclination and magnetic intensity of each specimen were recorded on the ScT following each demagnetisation step. Measurement of χ between heating steps was undertaken to check for possible heating-induced mineralogical changes. Results are analysed using principal-component analysis on Zijiderveld diagrams (Kirschvink 1980), utilising in-house software developed at the Black Mountain Paleomagnetic Laboratory by John Giddings.

RESULTS

NRM directions from the McKinnons samples are distributed around the modern pole position. Demagnetisation curves were multicomponent, with usually only one easily definable component (Figure 3), which displayed both normal and reversed directions, and were stable to high temperatures. A stereoplot of the characteristic remanent magnetisation (ChRM) directions highlights the two different populations (Figure 4). Thermal demagnetisation results for specimens from one block (MCK23) were not analysed, because a large increase in susceptibility indicated formation of a heating-induced magnetisation. No other specimens displayed this behaviour.

ChRM directions and calculated paleomagnetic poles are presented in Table 1. The bulk of the samples showed a ChRM with mixed polarity (44 normal and 38 reversed specimens), which were generally stable from about 200 to 700°C. These samples were from the bleached saprolite and mottled zones, including the transition from the bleached to ferruginous zones. The direction recorded by this ChRM had a declination of 011.9° and an inclination of –61.4°, with an α_{95} of 1.84°, which yields a paleomagnetic pole at 109.6°E, 75.2°S (A_{95} 2.50°). This plots directly to the west to northwest of the 16 Ma pole on the AAPWP of Schmidt & Clark (2000), with partially overlapping circles of A_{95} (Figure 5), and is assigned an age of 15 $\pm$ 4 Ma.

The other mean direction from the McKinnons mine samples was obtained from 35 reverse-polarity directions from the deeply oxidised zone (MCK06, 07, 15, 16, 17). All ChRMs were stable from 200 to 690°C. The mean direction obtained from these samples was defined by a declination of 205.3° and an inclination of 70.0°, with an

Table 1 Results from the McKinnons mine paleomagnetic study.

Site	Location	Component[a]	n (r)[b]	Remanence directions[c]				South Pole[c]				Age (Ma)
				Decl	Incl	k	α_{95}	Long	Lat	K	A_{95}	
Bleached saprolite and mottled zones												
01	West wall, 2nd berm (30 m)	HT	7(1)	002.5	–63.3	258	3.76	138.1E	76.8S	113	5.71	–
02	West wall, 2nd berm (30 m)	HT	7(4)	004.5	–63.7	73.8	7.07	131.0E	75.1S	36.7	10.1	–
03	South wall, 2nd berm (30 m)	HT	7(4)	018.8	–54.4	164	4.73	072.6E	74.2S	105	5.90	–
04	South wall, 2nd berm (30 m)	HT	6(0)	014.3	–58.8	70.2	8.05	094.8E	75.4S	40.3	10.7	–
08	Megamottle zone (7 m)	HT	7(0)	004.7	–64.7	198	4.31	132.9E	74.5S	86.6	6.52	–
09	Megamottle zone (7 m)	HT	5(5)	190.7	63.5	202	5.38	117.2E	74.1S	117	7.11	–
10	Southeast wall (13 m)	LT	5(0)	019.1	–56.2	58.1	10.1	079.2E	73.6S	32.4	13.7	–
	Southeast wall (13 m)	HT	3(1)	044.8	–72.8	62.2	15.8	112.5E	49.6S	28.3	23.6	–
	Southeast wall (13 m)	HT	2(0)	352.9	–55.7	54.7	34.5	192.0E	82.3S	51.2	35.6	–
11	Southeast wall (13 m)	HT	5(5)	019.1	–59.9	19.8	17.6	095.1E	70.6S	11.9	23.1	–
19	Southeast wall (25 m)	HT	7(7)	203.2	58.2	121	5.51	083.9E	69.8S	79.5	6.81	–
21	South wall (45 m)	HT	7(6)	193.9	64.5	403	3.01	113.5E	71.8S	195	4.33	–
22	South wall (45 m)	HT	7(4)	186.2	62.9	105	5.91	125.2E	76.1S	44.4	9.16	–
23	South wall; sulfide pod (45 m)	LF	4(0)	204.8	–9.7	38.1	15.1	003.0E	46.4S	41.9	14.4	–
Intensely weathered and oxidised zone												
05	Southeast wall (25 m)	HT	7(0)	013.3	–54.9	424	2.93	078.1E	78.4S	322	3.37	–
06	Southeast wall (25 m)	HT	7(7)	195.5	73.4	1035	1.88	129.9E	59.8S	366	3.16	–
07	Southeast wall (25 m)	HT	7(7)	205.6	67.6	482	2.75	108.3E	63.4S	237	3.93	–
15	Southeast wall (25 m)	HT	7(7)	206.7	69.7	2511	1.20	112.1E	61.1S	936	1.97	–
16	Southeast wall (25 m)	HT	7(7)	211.5	65.2	507	2.68	097.9E	61.5S	246	3.85	–
17	Southeast wall (25 m)	HT	7(7)	204.0	73.8	933	1.98	123.4E	57.6S	337	3.29	–
18	Southeast wall (25 m)	HT	7(3)	016.0	–64.5	55.5	8.17	111.0E	70.2S	29.8	11.3	–
Mean: all sites		HT	67(35)	012.0	–62.5	68.0	2.12	112.9E	73.9S	38.4	2.84	–
Mean: all sites excl. 06, 07, 15, 16, 17		HT	82(38)	011.9	–61.4	73.2	1.84	109.6E	75.2S	40.2	2.50	15 $\pm$ 4
Mean: sites 06, 07, 15, 16, 17		HT	35(35)	205.3	70.0	297	1.41	114.6E	61.3S	128	2.15	60 $\pm$ 10

[a]LT, low-temperature component; IT, intermediate temperature component (<580°C); HT, high-temperature component (>580°C).
[b]n, number of specimens; (r), number of specimens with reversed direction.
[c]k and K are Fisher precision parameters; α_{95} and A_{95} are semi-angles of 95% confidence.

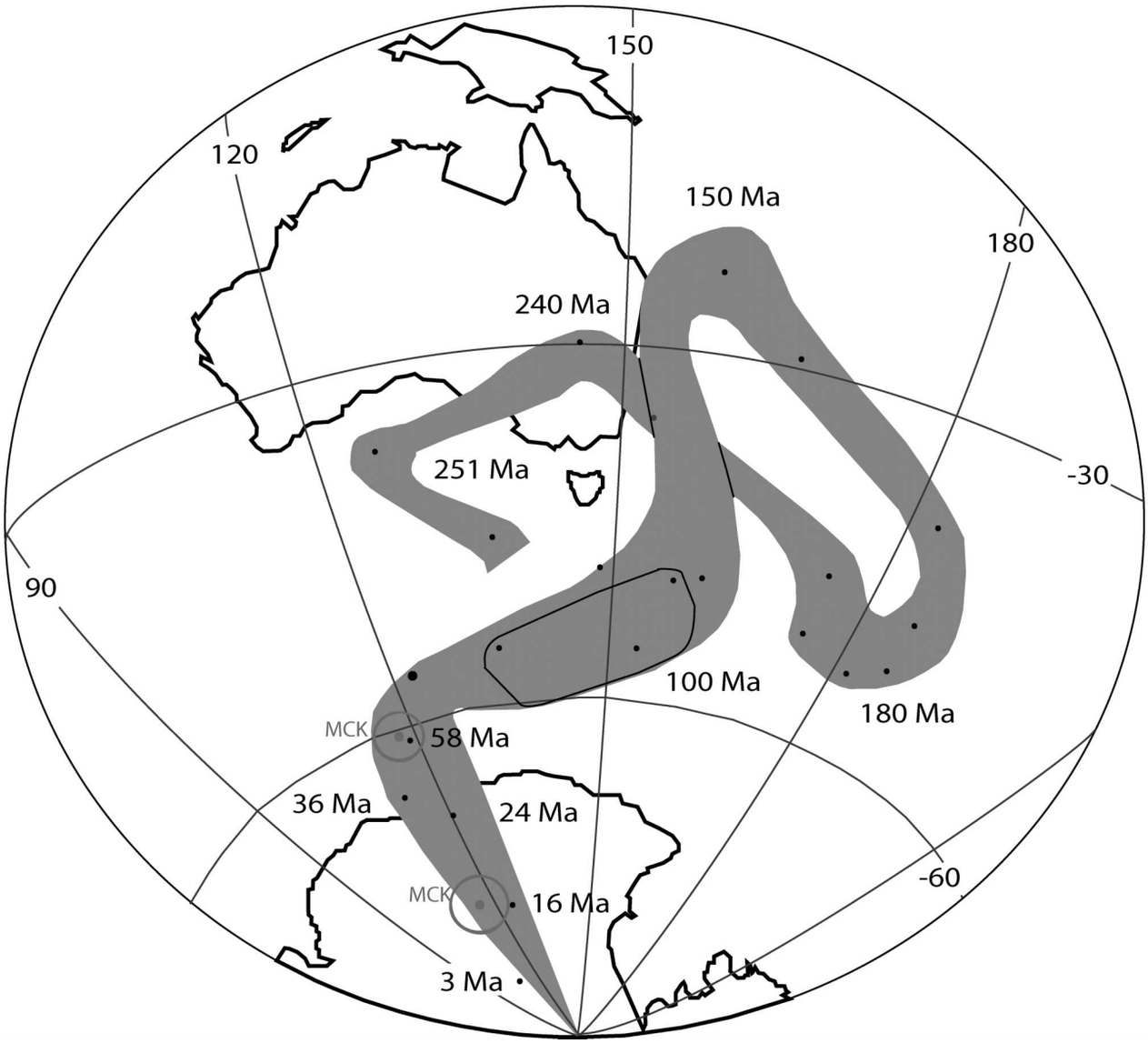


Figure 5 McKinnons pole positions plotted on the Australian apparent polar wander path (after Schmidt & Clark 2000). Reference pole positions shown in dots, McKinnons results shown with A_{95} circles.

α_{95} of 1.41°, yielding a pole at 114.6°E, 61.3°S (A_{95} 2.15°), which plots on the AAPWP, to the northwest of the 58 Ma reference pole (Figure 5), yielding an age of 60 ± 10 Ma.

IMPLICATIONS FOR REGIONAL REGOLITH DEVELOPMENT AND LANDSCAPE EVOLUTION

Traditional models of landscape evolution for central and western New South Wales have emphasised peneplanation or pediplanation concepts of classical geomorphology (King 1950; Langford-Smith and Dury 1965; Dury 1966; Ongley 1970). Some of these models also implied that duricrust layers (silcretes and indurated ferruginous zones) could be used as continuous morphostratigraphic markers. However, more recent studies have focused on the significance of drainage evolution and changes in base-level in the surrounding Mesozoic and Cenozoic sedimentary basins (Spry 1999, 2003; Gibson & Chan 1999). These later studies have found evidence for a variety of differently aged features and similar regolith materials preserved in various parts and levels of the landscape. This is consistent with a landscape evolution model incorporating continuous and concomitant weathering, erosion and deposition, primarily controlled by bedrock lithology and structure, climate and, to a lesser degree, tectonics. Ferruginised and silicified regolith materials, which are resistant to further chemical weathering and erosion, have also exerted a significant lithological control on landscape development. Parts of the Cobar region were inundated by the Albian marine transgression into the Eromanga Basin (Exon & Senior 1976; Frakes *et al.* 1987; Spry 1999; Gibson & Chan 1999). Since that time the region is thought to have been subaerially exposed. Incised

Downloaded By: [University Of Western Australia] At: 12:50 25 March 2009

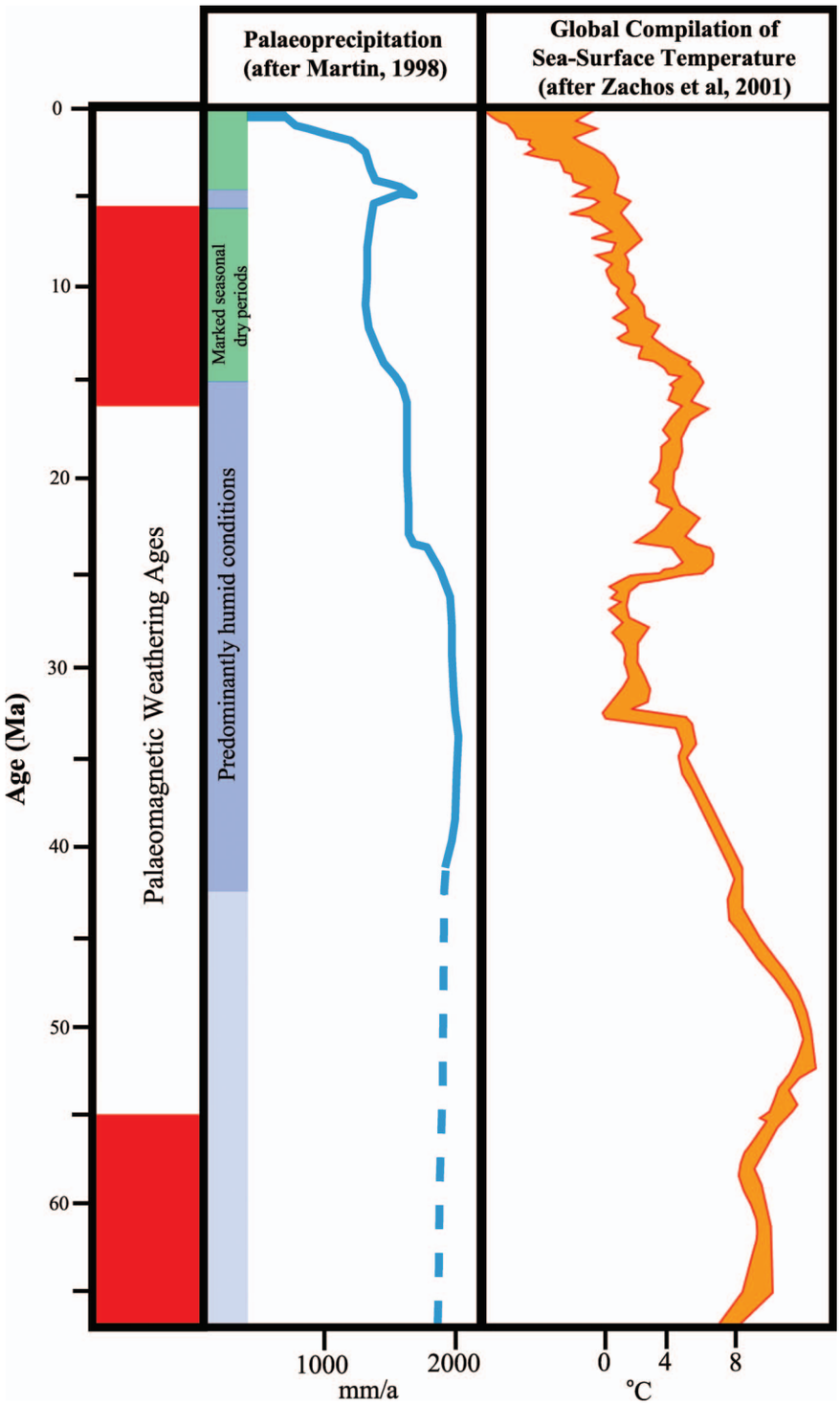


Figure 6 Comparison of paleomagnetic weathering ages from McKinnons mine with climate data from Martin (1998) and a global sea-surface temperature curve from Zachos *et al*. (2001).

paleovalleys that formed during periods of low base-level in the Cenozoic were later infilled by eroded regolith.

The results of the present study clearly demonstrate that ferruginous regolith at a single site has formed at different times. They also confirm subaerial exposure of the McKinnons area during the Cenozoic by demonstrating two episodes of intense oxidative weathering with formation of secondary hematite that had reached terminal stages in the Paleocene (*ca* 60 Ma) and Miocene (*ca* 15 Ma). This is also consistent with broadly similar ages for intense ferruginisation at other sites in the Cobar region (McQueen *et al*. 2007). These episodes of strong ferruginisation reflect significant changes in weathering, hydrological and chemical conditions developed during ongoing weathering through the Cenozoic. There is good stratigraphic control on the later episode of strong ferruginisation at a site near Wilga Tank, northeast of Cobar, where Miocene leucitite

Downloaded By: [University Of Western Australia] At: 12:50 25 March 2009

flows were erupted onto a low relief landscape with deep, oxidised and ferruginous weathering profiles, which are now preserved beneath the lava flows. Post-eruption weathering of the lavas has been minimal (McQueen *et al.* 2007).

WEATHERING AGES AND LONG-TERM CLIMATE CHANGE

The most important proxy for climate information spanning the entire Cenozoic is arguably the oxygen isotope ($\delta^{18}O$) record from deep-sea sediments (Kennett & Stott 1991; Bralower *et al.* 1995; Zachos *et al.* 1997). The $\delta^{18}O$ composition of marine benthic foraminifera is a first-order measure of global ice volume, from which it can be deduced that the general trend in global climate during the Cenozoic is towards cooler temperatures, driven largely by growth of the Antarctic ice sheet (Figure 6).

From $\delta^{18}O$ records and other evidence (e.g. paleobotanical evidence), it is clear that the southern hemisphere experienced a warmer, wetter climate throughout much of the Paleocene and Early Eocene than at present. Two major events are recognised within this interval, the first being a very warm period at the Paleocene/Eocene boundary (the Late Paleocene Thermal Maximum) and the second being the Early Eocene Climate Optimum. This so-called 'Greenhouse World' in the Early Cenozoic was brought to an end by initiation of the Antarctic ice sheet and rapid cooling of the Southern Ocean by $\sim 5^{\circ}C$ at around the Oligocene/Eocene boundary (Kennett *et al.* 1974; Kennett & Shackleton 1976; Savin 1977; Zachos *et al.* 2001), leading into the icehouse world of the Late Cenozoic.

Whereas today much of Australia is arid to semiarid, with low rainfall across much of the rest of the continent, Early Cenozoic Australia experienced less seasonal, warmer, wetter climates than today. As the Cenozoic progressed, the climate became more seasonal and temperate, with the onset of major aridity around 5–4 Ma (Bowler 1976), but intensifying from 3 to 2 Ma (Fujioka *et al.* 2005). Evidence from the Australian arid zone indicates that throughout much of the Cenozoic it experienced a much more humid climate than that prevailing since the start of the Quaternary. Evidence from the large internal drainage basins that enclose Lakes Eyre and Frome in central Australia indicates extensive lacustrine environments during the Miocene, with macrofossil evidence confirming the prevalence of subtropical conditions at the time (Alley 1998).

Martin (1998) examined the Cenozoic climate history of southeastern Australia, including southern New South Wales and Victoria, in the context of increasing aridity. From paleobotanical evidence, it was concluded that precipitation was >1500 mm during the Paleogene, declining from the Middle–Late Miocene to the Late Pliocene. There was a further decrease to 500–800 mm in the Plio-Pleistocene. Throughout the Cenozoic, grassland became more dominant, developing first in the northwest of Australia, and later in the southeast. Aridity in the central part of the continent, and seasonality closer to the coast, became predominant climate trends from the Miocene, and both have intensified since the Late Pleistocene. A generalised precipitation curve from Martin (1998) is shown in Figure 6, compared with a global sea-surface temperature curve derived from $\delta^{18}O$ records from deep-sea cores (Zachos *et al.* 2001).

Paleomagnetic ages obtained from the McKinnons mine equate well with the climatic record. Periods of drying out of the profile, associated with a lowering of the water-table, are marked by intense oxidation and hematite fixation, yielding a strong paleomagnetic signal. This appears to have occurred in the Paleocene resulting in a preserved magnetic age. The marked seasonality during the Miocene, shown in both the paleoprecipitation and paleotemperature records, was again favourable to the formation of ferruginous weathering profiles, as recorded by the Miocene age in the McKinnons profile. The lack of paleomagnetic ages between the Paleocene and the Miocene may be explained by limited changes in oxidation or hydrologic conditions, or may possibly reflect periods of greater landscape instability. Unstable, eroding land surfaces are not conducive to the preservation of paleomagnetic signals. While climate may be the main driver, regional tectonism and base-level change associated with transgressions and regressions in the Murray Basin will have had an added effect on the preservation of weathering profiles.

CONCLUSIONS

Paleomagnetic dating shows that during the Cenozoic, there were two major periods of secondary hematite formation at the McKinnons mine, in the Cobar region. The first of these reached its terminal stages in the Paleocene, and the second in the Middle Miocene, broadly consistent with other paleomagnetic data, which indicate continent-wide weathering regimes conducive to the formation of secondary hematite during these time periods (Pillans 2002). It appears that the Miocene weathering extended deeper than, or completely overprinted, the Paleocene weathering. A more favourable climate, more stable landscape, or both, conducive to this weathering regime, existed for a longer period in the Miocene than in the Paleocene. Erosion has resulted in the truncation of the Paleocene weathering profile. Reduced erosion, and a deepening weathering front, by the Early to Middle Miocene allowed preservation of part of the older profile and deeper weathering to produce the 15 Ma profile.

Results from this study indicate that the landscape evolution and weathering history of the region dates back at least to the Cretaceous–Paleocene boundary. This implies that the area was subaerially exposed for periods of ~ 10 Ma during the Paleocene–Eocene and the Miocene. The long-term landscape stability necessary for such deep oxidative weathering profiles to develop indicates that the Cobar region probably separated marine sedimentation in the Murray Basin from the Eromanga and Surat Basins. The ages provide further evidence for episodic deep, intense chemical weathering under favourable climatic conditions across the region, and substantiate other paleomagnetic ages, which

Downloaded By: [University Of Western Australia] At: 12:50 25 March 2009

cluster at *ca* 60 ± 10 Ma and 16 ± 10 Ma, throughout Australia (Idnurm & Senior 1978).

These weathering ages can be compared with reconstructions of Australian climates from previously published work, which show a cooling trend over the last 40 Ma, following an extended period of high mean annual temperatures in the Paleocene and Eocene. In conjunction with this cooling, total precipitation decreased, and rainfall became more seasonal. The weathering ages relate to the onset of seasonal climate and the initiation of aridity in the Late Miocene.

Recognition of two major periods of weathering-related hematite fixation, reflecting significant changes in paleoclimate through the Cenozoic, has important implications for geochemical dispersion in the regolith of the Cobar region. Iron oxides/oxyhydroxides, including hematite, are important hosts for elements released and dispersed from weathering ore deposits. Periods of humid, deep weathering would have dispersed some elements but fixed others in the iron oxides and oxyhydroxides. Subsequent weathering under drier or arid conditions would have imposed a chemically and physically different weathering regime with different dispersion processes, predominantly mechanical. These processes are discussed by other workers, and are reflected in the development of extensive sheetwash and surface lag (Scott *et al*. 1991; Cohen *et al.* 1996 and references therein).

ACKNOWLEDGEMENTS

This study was supported by the Australian Government's Cooperative Research Centre program within CRC LEME and also by the Australian National University. We thank Troy Resources for allowing access to the McKinnons mine site, and Colin Pain and Keith Scott for helpful reviews. MLS would like to thank Dave Cohen at the University of New South Wales for supervision during part of the project. Comments by two anonymous reviewers are also gratefully acknowledged.

REFERENCES

Acton G. D. & Kettles W. A. 1996. Geologic and paleomagnetic constraints on the formation of weathered profiles near Inverell, Eastern Australia. *Palaeogeography, Palaeoclimatology, Palaeoecology* **126**, 211–225.

Alley N. F. 1998. Cainozoic stratigraphy, palaeoenvironments and geological evolution of the Lake Eyre Basin. *Palaeogeography, Palaeoclimatology, Palaeoecology* **144**, 239–263.

Bembrick C. S. 1975. Murray Basin. *In*: Markham N. L. & Basden H. eds. *The mineral deposits of New South Wales*, pp. 555–568. Geological Survey of New South Wales, Sydney.

Bird M. I. 1988. An isotopic study of the Australian regolith. PhD thesis, Australian National University, Canberra (unpubl.).

Bird M. I. & Chivas A. R. 1989. Stable-isotope geochronology of the Australian regolith. *Geochimica et Cosmochimica Acta* **53**, 3239–3256.

Bishop P., Hunt P. A. & Schmidt P. W. 1982. Limits to the age of the Lapstone Monocline, N. S.W—a paleomagnetic study. *Journal of the Geological Society of Australia* **29**, 279–310.

Bowler J. M. 1976. Aridity in Australia: age, origins and expression in aeolian landforms and sediments. *Earth-Science Reviews* **12**, 279–310.

Bralower T. J., Zachos J. C., Thomas E., Parrow M., Paull C. K., Kelly D. C., Silva I. P., Sliter W. V. & Kyger C. 1995. Late Paleocene to Eocene paleoceanography of the equatorial Pacific Ocean: stable isotopes recorded at Ocean Drilling Program Site 865, Allison Guyot. *Paleoceanography* **10**, 841–865.

Brown C. M. & Stephenson A. E. 1991. Geology of the Murray Basin, southeastern Australia. *Bureau of Mineral Resources Bulletin* **235**.

Butler R. F. 1992. *Paleomagnetism: magnetic domains to geologic terranes.* Blackwell, Boston.

Byrnes J. G. 1993. *Background notes on geology and mineral deposits of the Cobar 1:250000 sheet.* New South Wales Geological Survey, Sydney.

Cohen D. R., Rutherford N. F., Dunlop A. C. & Alipour S. 1996. Geochemical exploration in the Cobar region. *In*: Cook W. G., Ford A. J. H., McDermott J. J., Standish P. N., Stegman C. L. & Stegman T. M. eds. *The Cobar Mineral Field—A 1996 Perspective*, pp. 125–155. Australasian Institute of Mining and Metallurgy, Melbourne.

Dammer D., Chivas A. R. & McDougall I. 1996. Isotopic dating of supergene manganese oxides from the Groote Eylandt deposit, Northern Territory, Australia. *Economic Geology* **91**, 396–401.

Duk-Rodkin A., Chan R. A. & 2004. Drainage evolution of the Cobar region. *In*: McQueen K. G. & Scott K. M. eds. *Exploration field workshop Cobar region 2004, Proceedings*, pp. 37–42. Cooperative Research Centre for Landscape Evolution and Mineral Exploration, Perth.

Dury G. H. 1966. Duricrusted residuals on the Barrier and Cobar pediplains of New South Wales. *Journal of the Geological Society of Australia* **13**, 200–307.

Elliott S. M., Bywater A. & Johnston C. 1998. McKinnons gold deposit, Cobar. *In*: Berkman D. A. & Mackenzie D. H. eds. *Geology of Australian and Papua New Guinean mineral deposits.* pp. 567–574. Australasian Institute of Mining and Metallurgy, Melbourne.

Exon N. F. & Senior B. R. 1976. The Cretaceous of the Eromanga and Surat Basins. *BMR Journal of Australian Geology & Geophysics* **1**, 33–50.

Frakes L. A., Burger D., Apthorpe M., Wiseman J., Dettmann M., Alley N., Flint R., Gravestock D., Ludbrook N., Backhouse J., Swarko S., Scheibnerova V., McMinn A., Moore P. S., Bolton B. S., Douglas J. G., Christ R., Wade M., Molnar R. E., McGowran B., Balme B. E. & Day R. A. 1987. Australian Cretaceous shorelines, stage by stage. *Palaeogeography, Palaeoclimatology, Palaeoecology* **59**, 31–48.

Fujioka T., Chappell J., Honda M., Yatsevich I., Fifield K. & Fabel D. 2005. Global cooling initiated stony deserts in central Australia 2–4 Ma, dated by cosmogenic ^{21}Ne–^{10}Be. *Geology* **33**, 993–996.

Gibson D. L. & Chan R. 1999. Aspects of palaeodrainage in the north Lachlan Fold Belt region. *In*: Taylor G. & Pain C. eds. *Regolith '98: new approaches to an old continent. Proceedings of the 3rd Australian Regolith Conference*, pp. 23–37. Cooperative Research Centre for Landscape Evolution and Mineral Exploration, Perth.

Glen R. A. 1987. *Geology of the Wrightville 1:100 000 sheet 8034.* Geological Survey of New South Wales, Sydney.

Glen R. A., Clare A. & Spencer R. 1996. Extrapolating the Cobar Basin model to the regional scale: Devonian basin-formation and inversion in New South Wales. *In*: Cook W. G., Ford A. J. H., McDermott J. J., Standish P. N., Stegman C. L. & Stegman T. M. eds. *The Cobar Mineral Field—A 1996 Perspective*, pp. 43–83. Australasian Institute of Mining and Metallurgy, Melbourne.

Glen R. A., MacRae G. P., Pogson D. J., Scheibner E., Agostini A. & Sherwin L. 1985. *Summary of the Geology and Controls of Mineralization in the Cobar Region.* Geological Survey of New South Wales, Sydney.

Hawke J. M., Bourke D. J., Cramsie J. N. & McNevin A. A. 1975. Great Australian Basin. *In*: Markham N. L. & Basden H. eds. *The Mineral Deposits of New South Wales.* pp. 512–537. Geological Survey of New South Wales, Sydney.

Heim J. A, Vasconcelos P. M., Shuster D. L., Farley K. A. & Broadbent G. 2006. Dating paleochannel iron ore by (U–Th)/He analysis of supergene goethite, Hamersley Province, Australia. *Geology* **34**, 173–176.

Idnurm M. 1985. Late Mesozoic and Cenozoic palaeomagnetism of Australia I. A redetermined apparent polar wander path. *Geophysical Journal of the Royal Astronomical Society* **83**, 399–418.

Downloaded By: [University Of Western Australia] At: 12:50 25 March 2009

IDNURM M. 1994. New Late Eocene pole for Australia, time-averaging of remanence directions, and palaeogeographic reference systems. *Geophysical Journal International* **117**, 827–833.

IDNURM M. & SCHMIDT P. W. 1986. Paleomagnetic dating of weathered profiles. *Geological Survey of India Memoirs* **120**, 79–88.

IDNURM M. & SENIOR B. R. 1978. Paleomagnetic ages of Late Cretaceous and Tertiary weathered profiles in the Eromanga Basin, Queensland. *Palaeogeography, Palaeoclimatology, Palaeoecology* **24**, 263–277.

KENNETT J. P., HOUTZ R. E., ANDREWS P. B., EDWARDS A. R., GOSTIN V. A., HAJOS M., HAMPTON M. A., JENKINS D. G., MARGOLIS S. V., OVENSHINE A. T. & PERCH-NIELSEN K. 1974. Development of the Circum-Antarctic Current. *Science* **186**, 144–147.

KENNETT J. P. & SHACKLETON N. J. 1976. Oxygen isotopic evidence for the development of the psychrosphere 38 Myr ago. *Nature* **260**, 513.

KENNETT J. P. & STOTT L. D. 1991. Abrupt deep-sea warming, palaeoceanographic changes and benthic extinctions at the end of the Palaeocene. *Nature* **353**, 225.

KING L. C. 1950. The cyclic land-surfaces of Australia. A geomorphological summary. *Proceedings of the Royal Society of Victoria* **62**, 79–95.

KIRSCHVINK J. L. 1980. The least-squares line and plane and the analysis of paleomagnetic data. *Geophysical Journal of the Royal Astronomical Society* **62**, 699–718.

LANGFORD-SMITH T. & DURY G. H. 1965. Distribution, character and attitude of the duricrust in the northwest of New South Wales. *American Journal of Science* **263**, 170–190.

LUCK G. R. 1970. The palaeomagnetism of some Cambrian and Ordovician sediments from the Northern Territory, Australia. *Geophysical Journal of the Royal Astronomical Society* **20**, 31–39.

MARSHALL S. M. & SCOTT K. M. 1999. Supergene derived McKinnons gold deposit and associated features in the surrounding weathered rock. *In*: Taylor G. & Pain C. eds. *Regolith '98: New Approaches to an Old Continent. Proceedings of the 3rd Australian Regolith Conference*, pp. 149–154. Cooperative Research Centre for Landscape Evolution and Mineral Exploration, Perth.

MARTIN H. A. 1998. Tertiary climatic evolution and development of aridity in Australia. *Proceedings of the Linnean Society of New South Wales* **119**, 115–136.

MCELHINNY M. W. & EMBLETON B. J. J. 1974. Australian palaeomagnetism and the Phanerozoic plate tectonics of eastern Gondwanaland. *Tectonophysics* **22**, 1–29.

MCQUEEN K. G., GOZALEZ O. R., ROACH I. C., PILLANS B. J., DUNLAP W. J. & SMITH M. L. 2007. Landscape and regolith features related to Miocene leucitite lava flows, El Capitan northeast of Cobar, New South Wales. *Australian Journal of Earth Sciences* **54**, 1–17.

MCQUEEN K. G., RUGLESS C. S. & WILLIAMS R. E. 2005. McKinnons gold deposit, Cobar district, New South Wales. *In*: Butt C. R. M., Robertson I. D. M., Scott K. M. & Cornelius M. eds. *Regolith Expression of Australian Ore Systems*, pp. 282–284. Cooperative Research Centre for Landscape Evolution and Mineral Exploration, Perth.

NOTT J. F., IDNURM M. & YOUNG R. W. 1991. Sedimentology, weathering, age and geomorphological significance of Tertiary sediments on the far south coast of New South Wales. *Australian Journal of Earth Sciences* **38**, 357–373.

O'SULLIVAN P. B., GIBSON D. L., KOHN B. P., PILLANS B. & PAIN C. F. 2000. Long-term landscape evolution of the Northparkes region of the Lachlan Fold Belt, Australia: constraints from fission track and paleomagnetic data. *Journal of Geology* **108**, 1–16.

ONGLEY E. D. 1970. A palaeogeomorphic study on the Cobar pediplain, New South Wales. PhD thesis, University of Sydney, Sydney (unpubl.).

PIDGEON R. T., BRANDER T. & LIPPOLT H. J. 2004. Late Miocene (U+Th)–^{4}He ages of ferruginous nodules from lateritic duricrust, Darling Range, Western Australia. *Australian Journal of Earth Sciences* **51**, 901–909.

PILLANS B. 1998. *Regolith Dating Methods: a Guide to Numerical Dating Techniques.* Cooperative Research Centre for Landscape Evolution and Mineral Exploration, Perth.

PILLANS B. 2002. Climate-driven weathering episodes during the last 200 Ma in Southern Australia. *Geological Society of Australia Abstracts* **67**, 428.

PILLANS B. 2005. Geochronology of the Australian regolith. *In*: Anand R. R. & de Broekert P. eds. *Regolith Landscape Evolution Across Australia*, pp. 1–12. Cooperative Research Centre for Landscape Evolution and Mineral Exploration, Perth.

PILLANS B., TONUI E. & IDNURM M. 1999. Paleomagnetic dating of weathered regolith at Northparkes Mine, NSW. *In*: Taylor G. & Pain C. eds. *Regolith '98: New Approaches to an Old Continent. Proceedings of the 3rd Australian Regolith Conference*, pp. 237–242. Cooperative Research Centre for Landscape Evolution and Mineral Exploration, Perth.

PILLANS B. J. 1977. An Early Tertiary age for deep weathering at Bredbo, southern NSW. *Search* **8**, 81–83.

SAVIN S. M. 1977. The history of the Earth's surface temperature during the past 100 million years. *Annual Review of Earth and Planetary Sciences* **5**, 319–355.

SCHMIDT P. W. & CLARK D. A. 2000. Paleomagnetism, apparent polar-wander path, & paleolatitude. *In*: Veevers J. J. ed. *Billion Year Earth History of Australia and Neighbours in Gondwanaland.* pp. 12–17. GEMOC Press, Sydney.

SCHMIDT P. W., CURREY D. T. & OLLIER C. D. 1976. Sub-basaltic weathering, damsites, palaeomagnetism, and the age of lateritization. *Journal of the Geological Society of Australia* **23**, 367–370.

SCHMIDT P. W. & EMBLETON B. J. J. 1976. Paleomagnetic results from sediments of the Perth Basin, Western Australia, and their bearing on the timing of regional lateritisation. *Palaeogeography, Palaeoclimatology, Palaeoecology* **19**, 257–273.

SCHMIDT P. W. & OLLIER C. D. 1988. Paleomagnetic dating of Late Cretaceous to Early Tertiary weathering in New England, N.S.W., Australia. *Earth-Science Reviews* **25**, 363–371.

SCHMIDT P. W., TAYLOR G. & WALKER P. H. 1982. Paleomagnetic dating and stratigraphy of a Cainozoic lake near Cooma, N.S.W. *Journal of the Geological Society of Australia* **29**, 49–54.

SCOTT K. M., RABONE G., & CHAFFEE M. A. 1991. Weathering and its effect upon geochemical dispersion at the polymetallic Wagga Tank deposit, N.S.W., Australia. *Journal of Geochemical Exploration* **40**, 413–426.

SPRY M. J. 1999. Examining the landscape evolution models of the Cobar region, New South Wales, evidence from regolith–landform features. *In*: Taylor G. & Pain C. eds. *Regolith '98: New Approaches to an Old Continent. Proceedings of the 3rd Australian Regolith Conference*, pp. 251–262. Cooperative Research Centre for Landscape Evolution and Mineral Exploration. Perth.

SPRY M. J. 2003. The regolith and landscape evolution of a low relief landscape: Cobar, central New South Wales, Australia. PhD thesis, University of Canberra, Canberra (unpubl.).

TATE S. E., GREENE R. S. B., SCOTT K. M. & MCQUEEN K. G. 2007. Recognition and characterisation of the aeolian component in soils in the Girilambone Region, north western New South Wales, Australia. *Catena* **69**, 122–133.

VASCONCELOS P. M., BECKER T. A., RENNE P. R. & BRIMHALL G. H. 1992. Age and duration of weathering by ^{40}K–^{40}Ar and ^{40}Ar/^{39}Ar analysis of potassium–manganese oxides. *Science* **258**, 451–454.

ZACHOS J. C., FLOWER B. P. & PAUL H. 1997. Orbitally paced climate oscillations across the Oligocene/Miocene boundary. *Nature* **388**, 567.

ZACHOS J., PAGANI M., SLOAN L., THOMAS E. & BILLUPS K. 2001. Trends, rhythms, and aberrations in global climate 65 Ma to present. *Science* **292**, 686–693.

Received 28 September 2007; accepted 7 August 2008

Downloaded By: [University Of Western Australia] At: 12:50 25 March 2009